

Notes for ‘Applications of the Product Rule’

Examples and derivations:

1. In earlier notes we made progress in understanding the behaviour of the Gaussian curve

$$y = e^{-x^2}.$$

We observed that the y -intercept is $y(0) = e^0 = 1$, which becomes, in fact, the global maximum value for the associated function.

All the values of the function are positive, so the curve sits above the x -axis, which is a horizontal asymptote.

We used the Chain Rule to find the derivative

$$y' = -2xe^{-x^2}.$$

yielding the following sign diagram, indicating that the curve is increasing from the left and decreasing to the right of the origin:

x		0	
y'	$+$	0	$-$
	/		\

We now have the tools to go further and find the second derivative.

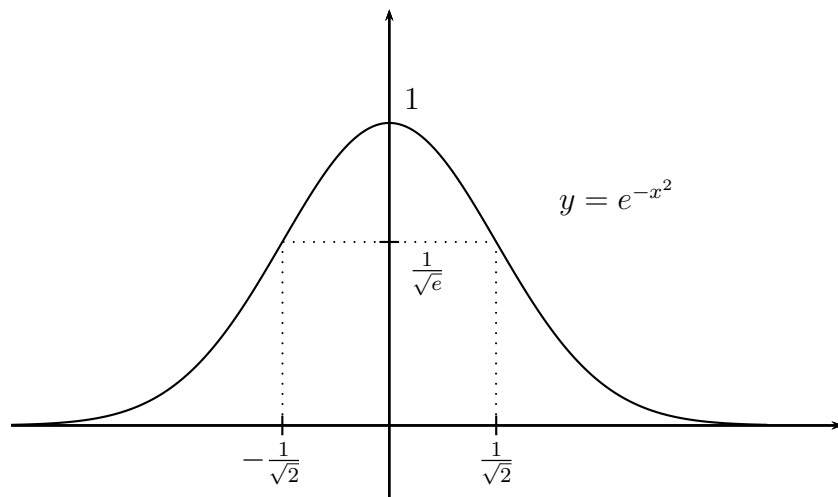
Observe that $y' = uy$ where $u = -2x$, so, by the Product Rule,

$$y'' = uy' + yu' = (-2x)(-2xe^{-x^2}) + e^{-x^2}(-2) = 2(2x^2 - 1)e^{-x^2}.$$

Hence $y'' = 0$ precisely when $2x^2 - 1 = 0$, that is, when $x = \pm \frac{1}{\sqrt{2}}$. Since e^{-x^2} is positive, the sign of y'' is completely determined by the sign of $2x^2 - 1$, and we get the following sign diagram:

x		$-\frac{1}{\sqrt{2}}$		$\frac{1}{\sqrt{2}}$	
y''	$+$	0	$-$	0	$+$
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As we suspected all along, the curve is concave up, followed by concave down, followed by concave up. The points of inflection occur when $x = \pm \frac{1}{\sqrt{2}}$.



2. We use the Product Rule to verify that, for all x and for all positive integers n , we have

$$\frac{d}{dx}(x^n) = nx^{n-1}.$$

Proof: We use a method known, more formally as *proof by mathematical induction*, which you may see explained in further courses in mathematics. You should be able to follow the main idea, even without a formal introduction to the general method.

The statement above is a claim about all positive integers, $n = 1, 2, 3, 4$, and so on, forever. We first check that it holds in the simplest case $n = 1$:

$$\frac{d}{dx}(x^1) = \frac{d}{dx}(x) = 1 = 1x^0 = 1x^{1-1},$$

which indeed is the case. Now suppose that the claim holds for some particular positive integer n , that is,

$$\frac{d}{dx}(x^n) = nx^{n-1}. \quad (1)$$

Now this may look like we are cheating, assuming what we want to prove, but that is not the case. We are only assuming the claim holds for one particular value of n , and then seeing if we can use this assumption to leverage up to other values of n .

Observe that $x^{n+1} = x^n(x) = uv$ where $u = x^n$ and $v = x$. Then $u' = nx^{n-1}$, by our assumption (1), and $v' = 1$. Hence, by the Product Rule,

$$\frac{d}{dx}(x^{n+1}) = uv' + vu' = x^n(1) + x(nx^{n-1}) = x^n + nx^n = (n+1)x^n,$$

that is,

$$\frac{d}{dx}(x^{n+1}) = (n+1)x^n. \quad (2)$$

We have carefully verified the following implication:

$$(1) \implies (2).$$

We checked earlier that (1) holds when $n = 1$, so this implication implies that (2) holds, which is statement (1) using $n = 1 + 1 = 2$.

But then the implication implies that (2) holds when $n = 2$, which becomes statement (1) again, but now with $2 + 1 = 3$.

Feeding back into the implication, we get (2) holding when $n = 3$, which is (1) again, but now with $n = 3 + 1 = 4$.

Feeding back into the implication, we get (2) holding when $n = 4$, which is (1) again, but now with $n = 4 + 1 = 5$.

We repeat this process of climbing up from one positive integer to the next, forever. By an “instantaneous” infinite chain-reaction, we therefore get that (1) holds for all positive integers n . This completes the proof.

3. We can go much further if we assume throughout that x is positive. We claim that, for any real number α ,

$$\frac{d}{dx}(x^\alpha) = \alpha x^{\alpha-1}.$$

The restriction that $x > 0$ ensures that the power x^α is always sensibly defined.

Proof: We make free use of properties of logs and exponentials throughout. Observe first that

$$x^\alpha = (e^{\ln x})^\alpha = e^{\alpha \ln x} = e^u,$$

where $u = \alpha \ln x$, so that

$$\frac{du}{dx} = \alpha \frac{d}{dx}(\ln x) = \frac{\alpha}{x}.$$

By the Chain Rule

$$\frac{d}{dx}(x^\alpha) = \frac{d}{dx}(e^u) = \frac{d}{du}(e^u) \frac{du}{dx} = e^u \left(\frac{\alpha}{x} \right) = \alpha x^\alpha x^{-1} = \alpha x^{\alpha-1},$$

completing the proof.